

## 1. Frontend Development (React)

### Project Structure

src/

```
|— components/  
|   |— About.jsx  
|   |— Projects.jsx  
|   |— Skills.jsx  
|   |— Resume.jsx  
|   |— Contact.jsx  
|— App.jsx  
|— index.js  
|— styles/
```

### Main App Component

```
import React from "react";  
import About from "../components/About";  
import Projects from "../components/Projects";  
import Skills from "../components/Skills";  
import Resume from "../components/Resume";  
import Contact from "../components/Contact";
```

```
function App() {  
  return (  
    <div>  
      <About />  
      <Projects />  
      <Skills />  
      <Resume />  
      <Contact />  
    </div>  
  );  
}
```

```
export default App;
```

### About Section

```
function About() {  
  return (  
    <section id="about">  
      <h1>About Me</h1>  
      <p>I am a passionate React developer with experience in building modern web apps.</p>  
    </section>  
  );  
}
```

export default About;



## Projects Section

```
function Projects() {  
  return (  
    <section id="projects">  
      <h1>Projects</h1>  
      <div className="project-card">  
        <h3>Portfolio Website</h3>  
        <p>Built using React and Tailwind CSS</p>  
      </div>  
    </section>  
  );  
}
```



## Skills Section

```
function Skills() {  
  return (  
    <section id="skills">  
      <h1>Skills</h1>  
      <ul>  
        <li>React</li>  
        <li>JavaScript</li>  
        <li>CSS</li>  
      </ul>  
    </section>  
  );  
}
```

```
}
```



## Resume Section

```
function Resume() {  
  return (  
    <section>  
      <h1>Resume</h1>  
      <a href="/resume.pdf" download>Download Resume</a>  
    </section>  
  );  
}
```



## Contact Section

```
function Contact() {  
  return (  
    <section>  
      <h1>Contact</h1>  
      <form>  
        <input type="text" placeholder="Name" />  
        <input type="email" placeholder="Email" />  
        <button type="submit">Send</button>  
      </form>  
    </section>  
  );  
}
```



## 2. Responsive Design

### CSS Example

```
.container {  
  display: flex;  
  gap: 20px;  
}  
  
@media (max-width: 768px) {  
  .container {  
    flex-direction: column;
```

```
}  
}
```

✓ Works on:

Mobile 📱

Tablet ➡ 📱

Desktop 💻

### ⚡ 3. Performance Optimization

#### ◆ Lazy Loading Components

```
import React, { Suspense } from "react";
```

```
const Projects = React.lazy(() => import("./Projects"));
```

```
function App() {  
  return (  
    <Suspense fallback={<div>Loading...</div>}>  
      <Projects />  
    </Suspense>  
  );  
}
```

#### ◆ Image Optimization

```

```

#### ◆ Minification (Automatic in build)

Removes unused CSS

Compresses JS

Improves load time

### 🏗️ 4. Production Build

Command

```
npm run build
```

Test Locally

```
npm run serve -s build
```

✓ Ensures:

All routes work

Assets load correctly

No console errors

## 5. Deployment

Option 1: Netlify

Steps:

Push code to GitHub

Go to Netlify

Import repository

Deploy

Option 2: Vercel

npm install -g vercel

vercel

Option 3: GitHub Pages

npm install gh-pages --save-dev

npm run deploy

## Custom Domain & SSL

Add domain in hosting dashboard

SSL enabled automatically (HTTPS)

## 6. Testing

### Checklist

Chrome, Firefox, Edge tested

Mobile responsiveness verified

All links working

Fast load speed

## 7. Cross-Browser Testing

Chrome 

Firefox 

Edge 

Safari (optional)

## 8. GitHub Submission

Repository Structure

portfolio/

|— src/

|— public/

|— README.md

|— package.json

 README.md (Use this in your repo)

# React Portfolio Website

## 🚀 Features

- Responsive design
- Lazy loading
- Optimized performance

## ⚙️ Tech Stack

- React
- CSS

## 📦 Installation

npm install

npm start

## 🏗️ Build

npm run build

## 🌐 Deployment

Deployed using Netlify/Vercel

## ⚠️ Challenges

- Responsive layout issues
- Fixed using Flexbox & media queries